Learning Whom to Trust from Feedback: Raw, Summarized and Stated Advisor Accuracy in Vision-Language Models

Anonymous submission

Abstract

[DRAFT: development episodes (16) and four models; replace with held-out results] Adapting to a volatile environment, as humans do when they learn whom to trust, is a hallmark of context-dependent learning. Learning whom to trust from feedback requires deciding on each trial which advisor was right, integrating these outcomes over trials, and using the estimate. We adapt a human advice-taking task with stable and volatile advisor accuracy to frozen vision-language models (VLMs) and give the same history in formats that remove these steps one at a time. With the raw history of reports and outcomes, models chose at or below chance; their choices were best described by repeating the previous outcome, and their weights on which advisor had been correct were close to zero. Stating on each trial who was correct raised accuracy by 15–17 percentage points, to the level of a summary of counts, whereas appending the same statement to the raw history recovered less than half of that gain. Models followed a summarized history almost exactly as the summary's own rule did, and used stated reliability almost perfectly, but unlike humans they did not weight recent outcomes more when accuracy was volatile. In these models, the bottleneck in learning advisor accuracy from experience is evaluating raw outcomes, not integrating them. **[TODO: finalize with held-out results and the human comparison; name volatility/context-dependent adaptation in the takeaway]**

Introduction

Deciding whom to trust is often learned from feedback. When two advisors disagree and the outcome is revealed afterwards, an observer can keep track of how often each advisor was right and weight their advice accordingly. Human participants do this in advice-taking tasks, and they revise their estimate of an advisor's accuracy faster when that accuracy is volatile (Behrens et al. 2008), as they do for reward probabilities (Behrens et al. 2007). In judgment and decision-making research, advisor reputations form quickly from a track record (Yaniv and Kleinberger 2000; Bonaccio and Dalal 2006). The problem also has well-studied normative solutions, from delta rules whose learning rate increases with volatility or hazard rate to Bayesian change-point models (Nassar et al. 2010; Mathys et al. 2011; Piray and Daw 2021). Because the advice-taking task comes from human neuroscience, its behavioral signatures, such as whether an observer fails at scoring outcomes or at integrating them and whether its learning rate rises when accuracy is volatile, can be measured in a model in the same way and compared with people.

Agents built on large language models (LLMs) and vision-language models (VLMs) increasingly face the same problem. An agent that combines tool outputs, retrieved documents or messages from other agents receives reports that conflict, and the reliability of their sources can change. Recent work has asked whether LLMs weight sources by reliability when reliability is fixed (Ashkinaze et al. 2026), when it flips once (Zhou et al. 2026), or after scripted failures (Chen 2026), and has proposed memories that track reliability outside the model (Feng, Yang, and Poria 2026). It remains open whether a frozen model (no parameter updates) can learn advisor accuracy in-context from trial-by-trial feedback when that accuracy changes over time, and, if it cannot, which part of the computation fails. This is context-dependent learning without plasticity: a person changes how much each new outcome moves their estimate when the environment becomes volatile, whereas a frozen model can adapt only through what its context contains.

Cognitive science offers a useful contrast. People choose differently when they are told outcome probabilities than when they learn them from experience, the description–experience gap (Hertwig et al. 2004; Wulff, Mergenthaler-Canseco, and Hertwig 2018). LLMs show an analogous description–history (DH) gap in risky choice (Ge, Zhang, and Vorobeychik 2026). In multi-armed bandits, the only LLM configuration that explored satisfactorily received a summarized history rather than a raw history (Krishnamurthy et al. 2024), and in a counting task, direct answers of open models were close to chance, with recency-weighted integration kernels (Xue, Wilson, and Xiong 2026). These studies show that the format of past evidence matters. They do not separate which processing step fails, they do not concern advisors, whose accuracy has to be inferred by comparing each report with the outcome, and they are text-only.

Learning advisor accuracy from a raw record involves three steps: (i) outcome evaluation, deciding on each trial

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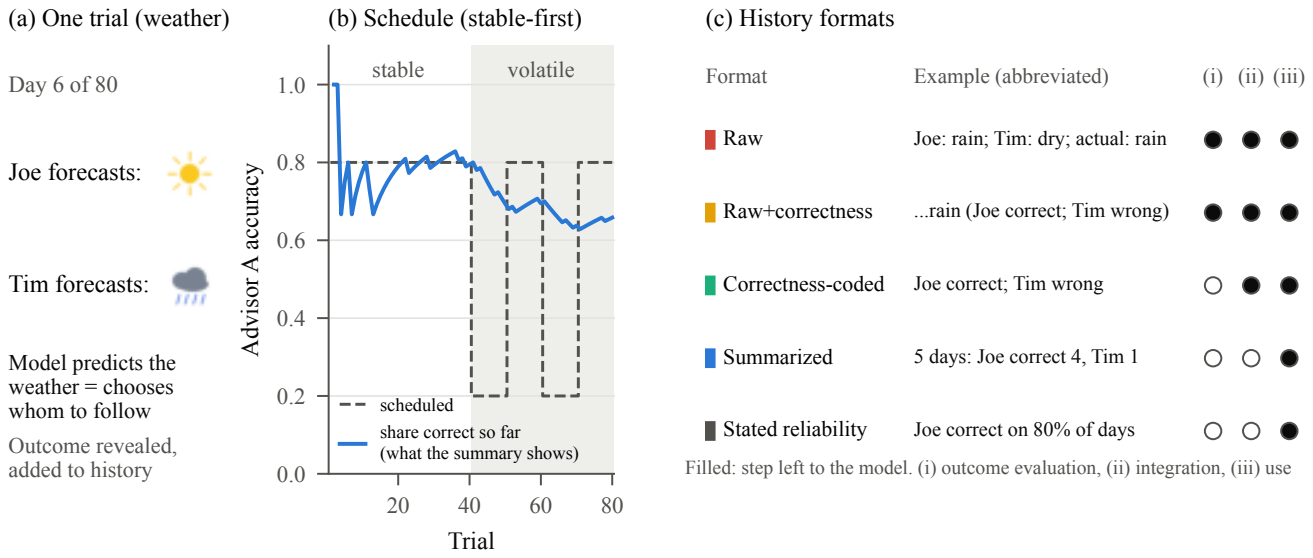


Figure 1: Task and history formats. (a) One trial in the weather cover story. (b) Scheduled accuracy of advisor A in a stable-first episode (dashed) and the share of past trials on which A was correct, which the all-trial summary reports (solid): it tracks the stable phase but lags behind every reversal. (c) History formats and the processing steps each leaves to the model: (i) outcome evaluation, (ii) integration across trials, (iii) use.

which advisor was correct; (ii) integration of these outcomes across trials; and (iii) use of the integrated estimate when choosing whose advice to follow. We adapt the stable and volatile advisor-accuracy schedule of Behrens et al. (2008) to frozen VLMs and present the same history in formats that remove these steps one at a time: the raw history, the raw history with correctness appended, a correctness-coded history in text or as a chart, and a summarized history of counts. In a separate stated-reliability condition, advisor accuracy is described directly. The comparison across formats indicates where performance is lost: a model that fails with the raw history but not with the correctness-coded one has difficulty with outcome evaluation, and one that fails with the correctness-coded history but not with the summary has difficulty with integration. Because the summary discards trial order, the comparison is informative only together with the correctness-coded formats and with control conditions that test each step on its own. The stable and volatile phases further allow us to ask whether a model, like human participants, weights recent outcomes more when accuracy is volatile. Reports are given as text or as images, including photographs, so the model must also attach each image to the right advisor. Our contributions are:

- A biologically grounded evaluation of context-dependent learning in VLMs: an advice-taking task adapted from a human neuroscience paradigm, with six text cover stories and one photo cover story, counterbalanced copies of every episode, and history formats that remove processing steps one at a time; the primary test was fixed before held-out evaluation.
- A test of where the gap between raw and summarized history arises, with integration kernels and strategy agreement to describe behavior under raw histories. In our development data the gap arises at outcome evaluation: models given raw histories behave as if repeating the previous outcome, whereas a correctness-coded history performs as well as a summary. Unlike human participants, models did not weight recent outcomes more when accuracy was volatile, even when outcome evaluation was done for them. **[TODO: confirm on held-out]**
- A supporting, descriptive comparison with human participants on an external task, the changing bandit in Psych-101 (Xiong et al. 2023), with models conditioned on the participants' previously executed behavior (Binz et al. 2025) and with the comparison's false-positive rate, bias and power calibrated by simulation.
- An open benchmark linking neuroscience and AI: the task generator with fixed seeds, all episodes and prompts, the control conditions, and the analysis and human-comparison pipelines. **[TODO: anonymous code link]**

Related Work

Cognitive evaluation of LLMs. Treating a language model as a participant in a psychological experiment is now a common way to characterize its behavior (Binz and Schulz 2023). CogBench turns such experiments into behavioral metrics (Coda-Forno et al. 2024), and Centaur was fine-tuned to predict trial-by-trial human choices in Psych-101 (Binz et al. 2025). Fits of reinforcement-learning models to in-context behavior in bandits have found asymmetric or biased updating and rigid or greedy choice (Schubert et al. 2024; Hayes, Yax, and Palminteri 2024; Wang, Ward, and

Zhang 2026b; Schmied et al. 2025). Wang, Ward, and Zhang (2026a) compared LLMs on reversal-learning schedules, and Reflection-Bench includes a reversal component (Li et al. 2025). In all of these tasks the model learns about its own options; it does not have to check another agent's report against an outcome.

Description, experience and history summaries. The description–experience gap (Hertwig et al. 2004) has an LLM counterpart in the DH gap of Ge, Zhang, and Vorobeychik (2026), where options are described either by explicit probabilities or by samples of past outcomes. Krishnamurthy et al. (2024) contrasted raw and summarized histories in stationary bandits, and Xue, Wilson, and Xiong (2026) traced integration kernels in a counting task, finding strong recency in direct answers that weakened when models were allowed to reason. Conversely, LLMs update roughly as Bayesian observers when the observed sequence is itself what they predict (Gupta et al. 2025), and behave like discounted Bayesian filters in non-stationary sequences (Zhang, Yang, and Wang 2025). Our summarized history follows Krishnamurthy et al. in being computed from the same history the model would otherwise see; it is neither lossless nor a description, and we keep it separate from the description condition D.

Volatility, change-points and multimodal reports. Our task follows the social-learning paradigm of Behrens et al. (2008). Normative accounts of learning under volatility include hierarchical Bayesian and change-point models (Behrens et al. 2007; Nassar et al. 2010, 2012; Mathys et al. 2011; Piray and Daw 2021, 2024; Soltani and Izquierdo 2019), and social and individual learning can be arbitrated separately (Diaconescu et al. 2020). We use Bayesian online change-point detection (BOCPD; Adams and MacKay 2007) only as an external integration rule, not as a model of the VLM. For image reports, performance depends on modality even when content is isomorphic (Fu et al. 2024), multi-image inputs show position biases (Tian et al. 2025), and VLMs have difficulty binding features to objects (Campbell et al. 2024).

Task and Methods

Advice-taking task. On every trial, two advisors give conflicting binary reports (for example, one forecasts rain and the other dry), the model predicts the outcome, and the outcome is then revealed (Fig. 1a). Because the reports always conflict, predicting the outcome amounts to choosing whose advice to follow. Outcome sequences are pre-generated with full feedback: both advisors' correctness is revealed on every trial, and the model's choices do not affect the subsequent history. An episode has 80 trials. We call the advisor favored on trial 1 advisor A. In the stable phase (40 trials), A is correct with probability 0.8; in the volatile phase (40 trials), A's accuracy alternates between 0.8 and 0.2 every 10 trials (Fig. 1b). The volatile phase follows the 80%-correct/80%-incorrect alternation of Behrens et al. (2008) (Supplementary Information); their stable phases differ from ours, so our schedule is adapted from theirs rather than identical. Half of the episodes run stable then volatile (stable-first) and half volatile then stable (volatile-first); both orders contain four reversals. Within each 10-trial block, A's realized accuracy equals the scheduled value exactly and the two outcome labels occur five times each (Appendix B).

Each episode is shown with two cover stories that share the same outcome sequence: a photo cover story, in which reporters say whether a hidden animal is a cat or a dog, and one of six text cover stories (colors, weather forecasts, stock predictions, sports predictions, medical readings, news claims), assigned in blocks so that cover story is crossed with phase order. The cover stories change only the wording, the answer words and the report images; they are not real-world datasets, and all outcome sequences are synthetic. Current reports are given either as text or as an image: a programmatically drawn icon without text for the text cover stories, and a photograph from the Oxford-IIIT Pet dataset (Parkhi et al. 2012) for the photo cover story. Every episode has a counterbalanced copy in which advisor names, answer labels and the order in which the instructions introduce the advisors are swapped while outcomes and images are identical. Within each trial, the order in which the two reports are listed is random. The instructions say that an advisor's accuracy may change over time ("A forecaster's accuracy may change over time.") but give no rate. There are 16 development and 48 held-out episodes (128 with counterbalanced copies); the held-out set has eight episodes per text cover story.

History formats. Before trial t, the model sees trials $1, \ldots, t-1$ in one of the following formats (Fig. 1c; full prompts in Appendix A). *Raw history*: each past trial lists both reports and the outcome ("Day 3: Ken forecasts rain; Sam forecasts dry. Actual weather: rain."); the model must carry out steps (i)–(iii). *Raw history with correctness*: the raw history with a statement of who was correct appended to each trial, which adds correctness information without removing the reports. *Correctness-coded history*: each past trial states only which advisor was correct ("Day 3: Ken was correct; Sam was wrong."), removing step (i). *Correctness chart*: the same information drawn as a grid image (green for correct, gray for wrong), read visually. *Summarized history*: one line with each advisor's number of correct trials so far ("Summary of all n past days: Ken was correct x times; Sam was correct y times."), removing steps (i) and (ii). The summary is computed from the same history and adds no observations, but it discards trial order, which matters when accuracy reverses; it is therefore not lossless, and a difference between summarized and raw history cannot be attributed to integration alone. Further summaries apply other integration rules outside the model: counts over the last K trials, over the segment since the last change-point inferred by Bayesian online change-point detection (BOCPD; Adams and MacKay 2007), or since the last true reversal (Appendix E).

Control conditions. On development episodes we run control conditions in which the answer is given more directly, to establish that each model can use reliability information. The most important is the stated-reliability condition, in which reliability is described in words ("Ken was correct on 80% of past days; Sam was correct on 20% of past days."); its contrast with the raw history is our measure of the description–history (DH) gap. Further controls state counts over the last

10 trials, give short (10-trial) or long (80-trial) histories in which advisor A is correct on every recent trial, in raw and correctness-coded versions, ask which advisor was correct on a single raw trial (step (i) alone), and ask which label a single report image shows (a perception check). Each control crosses all seven cover stories, text and image reports, and four query trials ($t \in \{20, 40, 60, 80\}$).

Models and readout. We test five frozen, instruction-tuned open-weight VLMs from two language-model families: Qwen3-VL 8B and 32B (Bai et al. 2025), InternVL3.5 8B and 38B (Wang et al. 2025) and Gemma-3-27B (Gemma Team 2025) (Table 1; details in Appendix K). InternVL3.5 uses a Qwen3 language model, so it adds a second vision pipeline but not a third family. Models run in bfloat16 with a float32 output projection. We read the logits of the first generated token for the two answer words, before any logit processing, and normalize them with a softmax over these two tokens to obtain y_t, the conditional preference between the two answer options (normalized over the two candidate tokens) for advisor A's current report. Because the normalization ignores all other tokens, y_t is neither the model's free-generation choice probability nor a belief. For every model, format and condition we therefore report the probability mass on the two candidate tokens (mean, minimum and share of trials below 0.5) and the format-following rate, the share of trials on which the greedy output is one of the two answer words (Appendix K). Decoding is greedy, and reasoning ("thinking") is switched off wherever the chat template offers it. On text-report trials with raw history we also ask a judgment question ("which forecaster is currently more likely to be correct?"). Reasoning is tested with one reasoning checkpoint on a subset of items (Appendix H).

Measures. Within each pair of an episode and its counterbalanced copy we average the label-aligned logit differences, which cancels additive name and label preferences. Accuracy is the expected proportion of trials ($t \geq 2$) on which the followed advisor was correct; it is 0.5 at chance and about 0.8 for an observer who knows the current accuracy. We also report the probability of following the currently more accurate advisor by phase, the log score of the outcome, and learning curves aligned to reversals. To describe behavior, we fit an integration kernel (cf. Xue, Wilson, and Xiong 2026): a logistic regression of the preference y_t on which advisor was correct at lags $k = 1, \ldots, 15$, plus a positive-recency regressor (choose the advisor whose current report matches the previous outcome), a label prior and a listing-order term. We also compute strategy agreement (Cohen's κ) between model choices and simple strategies, including positive recency, following the advisor correct on the previous trial (the CPT regressor of Behrens et al. 2008), counts over all or recent trials, and BOCPD. These analyses describe behavior that is consistent with a strategy; they do not identify a mechanism. To ask whether a model, like human participants, responds more strongly to recent outcomes when accuracy is volatile, we fit the kernel separately to trials of the stable and of the volatile phase and compare the summed weight of the last three outcomes (volatile/stable ratio, paired episode bootstrap; this analysis was specified after the analysis plan was fixed and is exploratory).

Analysis. The unit of analysis is the episode pair; all measures are averaged within episode first. The primary test, fixed before held-out evaluation (timestamped commit; not a formal preregistration), is the raw–summarized history gap: accuracy with the summarized history minus accuracy with the raw history, for text reports, per core model, tested with a two-sided Wilcoxon signed-rank test with Holm correction across core models; the decision criterion is a gap significantly above zero in both language-model families. We report the paired gap, the absolute accuracy under both formats, and the models and conditions in which the gap holds; accuracy with raw history is called near chance only if it lies within 0.5 ± 0.03. Secondary tests are two one-sided tests (TOST; Lakens 2017) of whether a VLM given a summary matches the accuracy of the summary rule alone, with equivalence bounds of ± 2 percentage points as the smallest effect size of interest; the position of the correctness-coded history between raw and summarized history, $(\mathrm{acc}_{\text{coded}} - \mathrm{acc}_{\text{raw}})/(\mathrm{acc}_{\text{summary}} - \mathrm{acc}_{\text{raw}})$, interpreted only when the summary–raw gap exceeds 0.05 and its CI excludes zero (the ratio need not lie in $[0, 1]$); and the primary test repeated on episodes with independent outcomes and irregular reversals. We report 95% bootstrap CIs over episode pairs (10,000 resamples), treat cover story as a random factor (per-story estimates and a sign test across the six text cover stories, whose smallest attainable two-sided p is .031), and use development episodes only for setting choices such as the window length K. Held-out episodes are run once.

Results

[DRAFT: development episodes (16) and four models; replace with held-out results]

Stated vs. experienced reliability (control conditions). All models used reliability when it was stated (Fig. 2b; Table 4 in Appendix C). With text reports, a stated percentage or stated counts led to the better advisor on 92–100% of items, and models identified which advisor had been correct on a single shown trial on 100% of items. The same information given as experience was used much less: with an easy raw history in which one advisor had been correct on every recent trial, accuracy was .60–.70 for 10-trial histories and .53–.56 for 80-trial histories. When the same easy histories were correctness-coded, accuracy rose to .996–1.00 for 10 trials and .68–.85 for 80 trials. Evaluating a single trial is therefore not the difficulty; evaluating and combining many raw trials is, and long histories are hard even when evaluation is removed.

Raw vs. summarized history. On the main task, accuracy with the raw history was .435–.487, at or below chance, and accuracy with the summarized history was .616–.632 (Table 1, Fig. 2a). The summary–raw gap was $+.129$ to $+.190$ in every model, including Gemma-3-27B from the second language-model family (16 development episode pairs; Wilcoxon $p < .001$ in each model). The correctness-coded history reached .608–.642, about as high as the summary: the coded history recovered 91–120% of the summary–raw

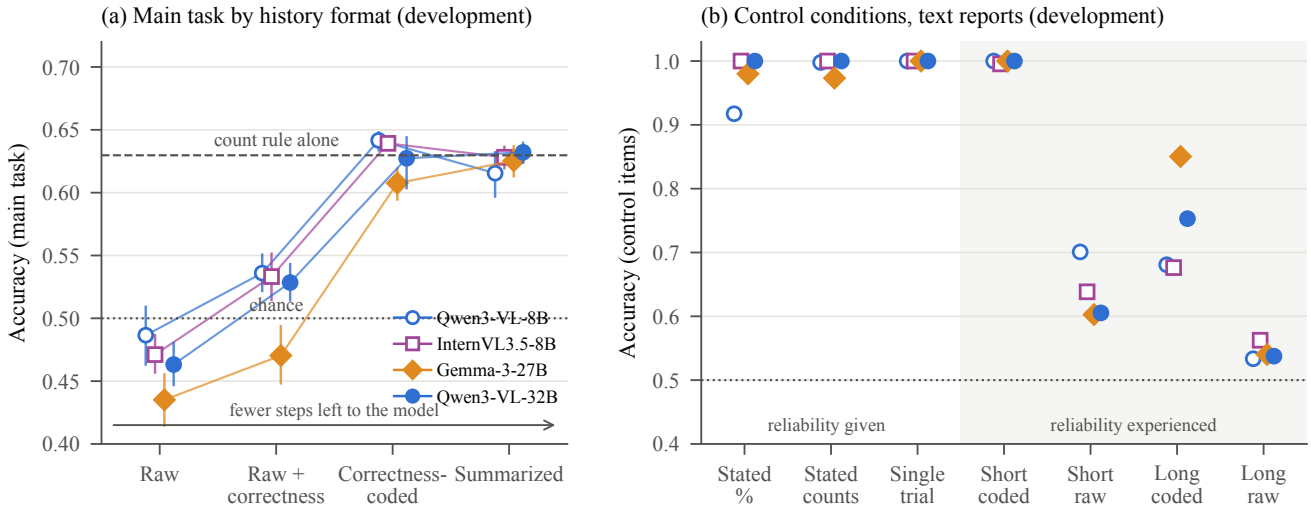


Figure 2: (a) Accuracy on the main task by history format, text reports; formats are ordered by how many processing steps they leave to the model. One line per model; marker shape and colour give the family (circle: Qwen3-VL; square: InternVL3.5; diamond: Gemma 3), filled markers the larger models; bars: 95% bootstrap CIs over episode pairs. Dashed: the all-trial count rule that produces the summary, applied without a model. (b) Control conditions with text reports, on their own metric (accuracy on control items). Stated: reliability given as a percentage or as counts. Single trial: who was correct on one shown trial. Short/long: easy 10- or 80-trial histories, raw or correctness-coded. **[DRAFT: development episodes (16) and four models; replace with held-out results]**

gap (Table 6). Appending the correctness statement to the raw history, so that the information was present but the reports remained, recovered only $+.035$ to $+.065$, one-fifth to two-fifths of the gap. In that history, replacing each past outcome label with a dash while keeping the correctness statement raised accuracy by 7–10 points, whereas replacing the reports did not; with both replaced, accuracy was about 2 points below the coded history (two 8B models; Fig. 3c, Appendix J). The visible outcome label, not the length of the record, keeps the appended statement from being used. The gap therefore arises mainly where the raw reports must be compared with the outcome, not where outcomes must be integrated. Integration over long histories still limits accuracy in the volatile phase, where following the better advisor was near .50 in all formats (Appendix C, Fig. 5), as it is for the all-trial count rule itself.

Behavior under raw histories. Under raw histories, the regression weights on which advisor had been correct were close to zero at every lag, whereas the weight on repeating the previous outcome (following whichever advisor currently reports the last trial's outcome) was positive in every model (0.28–0.78 log-odds; Fig. 3). Among simple strategies, repeating the previous outcome agreed best with the models' choices (Cohen's $\kappa = 0.24$–0.35; Table 11); it was the best-matching strategy in 11 to 14 of the 16 episodes for each model. This also explains accuracy below .5: because each outcome label occurs exactly five times in every 10-trial block (Appendix B), the previous outcome predicts the next one slightly worse than chance, and repeating it scores .439 on these episodes (.479 on the robustness set, whose outcomes are independent). With correctness-coded histories the kernel was recency-weighted (largest weight at lag 1) and the weight on the previous outcome vanished; with appended correctness both components were present. These analyses describe behavior consistent with a strategy and do not identify a mechanism.

Summaries and external integration rules. Given a summary of counts over all past trials, models chose the advisor with more correct trials on 92.6–99.4% of non-tied trials, and their accuracy was within ± 2 percentage points of the count rule alone in three of four models (TOST $p \leq .002$; Qwen3-VL-8B: $-.014$, 90% CI $[-.026, -.004]$, not equivalent; Table 10). A summary thus transfers the rule's accuracy to the model, including its weakness after reversals. **[TODO: change-point and window summaries (not yet run)]**

Multimodal reports. Image reports added a binding problem. Models named the content of a single report image almost perfectly (.99–1.00 for photos and icons), but with stated reliability and image reports, accuracy fell from .92–1.00 to .65–.84, and it was lowest when the images followed the advisor names rather than being labelled with them (Appendix G). With correctness-coded histories, accuracy fell from .608–.642 with text reports to .529–.553 with image reports. Models read a single cell of the correctness chart on only .60–.71 of items, below the prespecified gate of 0.8, so we do not interpret results with the chart history. These analyses are exploratory.

Adaptation to volatility. Human participants learn faster about an advisor when the advisor's accuracy is volatile

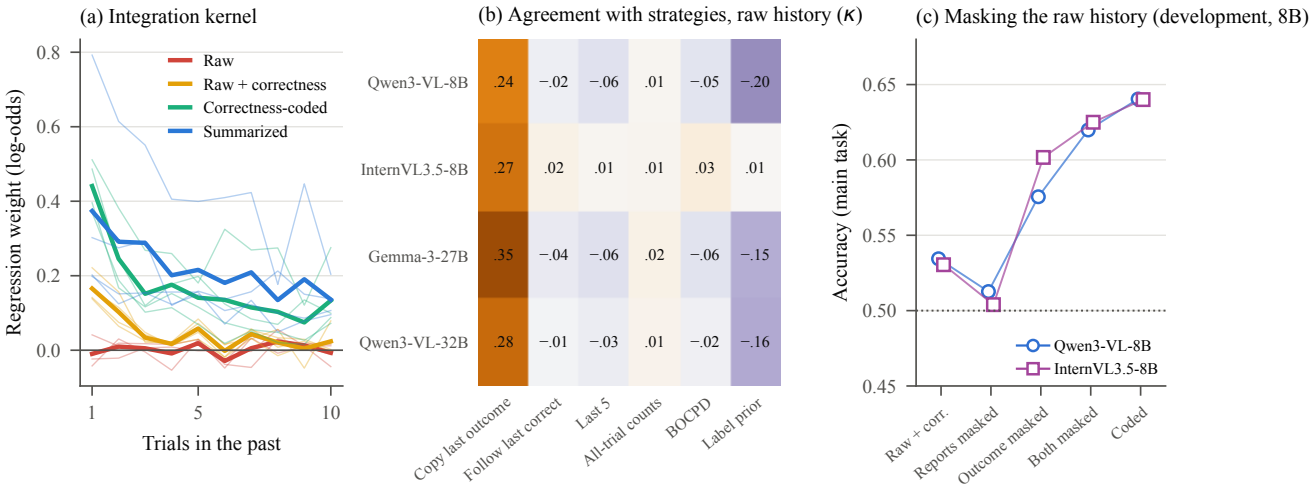


Figure 3: Behavior under raw histories. (a) Integration kernel: logistic-regression weight on which advisor was correct k trials in the past, by history format (thick: mean over models; thin: individual models; per-model kernels with CIs in Appendix F). Under raw histories the weights on past correctness are near zero. (b) Agreement (Cohen's κ) between model choices and simple strategies under raw histories; copying the previous outcome agrees best in every model. (c) Masking parts of the raw history with correctness: replacing the outcome label, but not the reports, recovers most of the difference to the coded history (Appendix J). **[DRAFT: development episodes (16) and four models; replace with held-out results]**

(Behrens et al. 2008). A learner with a constant learning rate, run through our phase-specific kernel analysis on the same episodes, gives a volatile-to-stable ratio of 0.96 for the summed weight of the last three outcomes, and a learner that speeds up under volatility gives a ratio above 1. With correctness-coded histories, where models do use past correctness, the ratio was 0.58–0.82, and with summarized histories 0.49–0.66: models weighted recent outcomes less, not more, when accuracy was volatile. Removing the evaluation step therefore restored the use of outcomes but not the context-dependent adjustment that people show; a summary does not provide it either, because the all-trial count lags behind every reversal (Fig. 1b) and models follow it. **[DRAFT: development episodes (16) and four models; replace with held-out results] [TODO: held-out ratios with CIs; calibrated effective learning rates (Appendix F) as a check]**

Comparison with humans. The changing bandit of Psych-101 tests the same signature in a second task with human data: a normative learner forgets the unchosen option faster when the hazard rate is higher, and participants did so (median forgetting rate ϕ from .079 at $h = 0.1$ to .255 at $h = 0.4$; Appendix I). This comparison is supporting and descriptive. Human participants were told the hazard rate and had played earlier games, whereas the models' primary condition hides it, so we report the paired difference between the model's and the human's slope of the forgetting rate ϕ on hazard rate (30 participants' histories) with a participant-bootstrap 95% CI, next to the comparison in which both are told the hazard rate; the human fits are our re-analysis. A simulation of the complete pipeline (Appendix I) gave a false-positive rate of 6.0% ($\pm$1.7%) when there is no true difference, CI coverage of 0.935, an upward bias of about $+0.09$ in the 20-draw estimator, and power of only 0.49–0.59 for a slope difference of ± 0.5. We therefore make no strong claim of a human–model difference or similarity, and a non-significant difference is not evidence of similarity. **[TODO: estimate and CI from held-out runs]**

Figure placeholder: Fig. 5. Forgetting rate ϕ vs. hazard rate: humans (told h; our re-analysis of Xiong et al. 2023) and models (hidden h; told h as secondary).

Figure 4: **[TODO: Human comparison on Psych-101.]**

Reasoning mode. **[TODO: Reasoning (runs pending).]**

Discussion and Limitations

[DRAFT: development episodes (16) and four models; replace with held-out results] The models in our study used stated reliability almost perfectly and followed a summary of counts as faithfully as the rule that produced it, yet chose at chance when they had to learn advisor accuracy from the raw record of reports and outcomes. The format comparison locates most of this loss at outcome evaluation: once each trial stated who was correct, accuracy matched the summary, even though integration over the history was still left

Model	LM	Main-task accuracy (held-out, 48 episodes)						Stated
		Raw	Raw+corr.	Coded	Chart	Summary	Summary−raw [95% CI]	reliability
Qwen3-VL-8B	Qwen3	.487	.536	.642	–	.616	+.129 [.105, .152]	.917
InternVL3.5-8B	Qwen3	.471	.533	.639	–	.628	+.157 [.141, .173]	1.000
Gemma-3-27B	Gemma 3	.435	.470	.608	–	.625	+.190 [.170, .211]	.980
Qwen3-VL-32B	Qwen3	.463	.528	.627	–	.632	+.169 [.152, .185]	1.000
InternVL3.5-38B	Qwen3	??	??	??	??	??	??	??
Rule alone (all-trial counts)		.630 (development) / .632 (held-out)					–	–

Table 1: **[DRAFT: development episodes (16) and four models; replace with held-out results]** Chart: not yet run. Accuracy on the main task by history format: proportion of trials ($t \geq 2$) on which the model follows the advisor who is correct, averaged over counterbalanced copies in logit space and then over episodes. LM: language-model family (InternVL3.5 uses a Qwen3 language model). Text reports; text and pet cover stories pooled (per cover story in Appendix J). Raw+corr.: raw history with correctness appended; Coded: correctness-coded history; Chart: correctness chart. Summary−raw: paired difference over episodes with 95% bootstrap CI. Stated reliability is a control condition run on development episodes with its own metric (accuracy on control items, Table 4), so it is not on the same scale as the main-task columns.

Model	Primary: summary − raw				Secondary: VLM+summary − rule alone (TOST, ±2 pp)					
					All trials		Change-point		Selected window	
	Est.	95% CI	p	p_{Holm}	Δ [90% CI]	p_{TOST}	Δ [90% CI]	p_{TOST}	Δ [90% CI]	p_{TOST}
Qwen3-VL-8B	??	??	??	??	??	??	??	??	??	??
InternVL3.5-8B	??	??	??	??	??	??	??	??	??	??
Gemma-3-27B	??	??	??	??	??	??	??	??	??	??
Qwen3-VL-32B	??	??	??	??	??	??	??	??	??	??
InternVL3.5-38B	??	??	??	??	??	??	??	??	??	??

Table 2: Primary and secondary tests on held-out episodes, fixed before held-out evaluation (timestamped commit); not a formal preregistration. Primary: Wilcoxon signed-rank test over episodes, Holm-corrected across core models; the claim requires a positive, significant gap in both language-model families (Qwen3 and Gemma 3). Secondary: two one-sided tests per model with Holm correction over the three summaries; the window length is selected on development episodes (Appendix E).

to the model. Under raw histories, choices were instead consistent with repeating the previous outcome, a strategy that ignores the advisors altogether. In the human task, the step the models fail at corresponds to computing a prediction error about the advisor, which Behrens et al. (2008) modelled separately from the reward prediction error and associated with a division of the anterior cingulate cortex distinct from the one tracking reward. We make no claim about neural mechanisms; at the computational level, the models fail at forming this error, not at integrating it once it is given. This differs from the human pattern in advice-taking tasks, where participants track advisor accuracy and adjust their learning rate to volatility (Behrens et al. 2008), and it refines the description–history gap (Ge, Zhang, and Vorobeychik 2026): here the gap between stated and experienced reliability is not mainly a failure to integrate experience, but a failure to turn raw experience into the outcomes that should be integrated. It also explains why summarized histories help in bandit tasks (Krishnamurthy et al. 2024): a summary performs the evaluation step for the model.

Removing the evaluation step was not enough for context-dependent learning. With correctness-coded or summarized histories, the models used past outcomes but did not respond more strongly to them when accuracy was volatile, whereas people raise their learning rate in that case (Behrens et al. 2007, 2008) and, in the changing bandit, forget faster when the hazard rate is higher. The models' in-context learning thus lacks both computations that the human literature separates: forming a prediction error about the advisor, and scaling updates by the estimated volatility.

Implications for NeuroAI. For building agents, the results suggest the division of labour that the human work describes: an explicit step that records which source was correct, analogous to an advisor prediction error, and an integration rule outside the model whose time scale is chosen for, or adapted to, the expected volatility (Appendix D); once a model follows a summary, the summary's rule determines how the agent responds to changes in reliability. For neuroscience, the format decomposition offers a way to test candidate models of advice taking step by step, against the same behavioral signatures as people; the frozen VLMs tested here are not yet such models. These are parallels at the computational level only; we make no claim about neural mechanisms. **[TODO: update with held-out results, window and change-point summaries, reasoning and human comparison]**

Limitations. The task is binary, with two advisors whose reports always conflict, and fixes advisor accuracy within 10-trial blocks; the fixed outcome quotas allow a quota-based shortcut, which is why we repeat the primary test on episodes

with independent outcomes and irregular reversals. Histories are text except for the correctness-coded chart, so images enter only as current reports. The photo-binding and chart-legibility checks use development episodes only and are exploratory. The main readout is an immediate single-token answer; reasoning was tested with one 8B reasoning model on a subset of items, and longer reasoning or larger models may change the picture (Xue, Wilson, and Xiong 2026). We tested two language-model families; InternVL3.5 shares the Qwen3 language model. The summarized history is not lossless, so the raw–summarized gap shows that the presentation of past evidence matters but does not by itself isolate integration; the correctness-coded formats and single-trial controls are needed for that step. Instructions mention that accuracy may change. The volatility analysis was specified after the analysis plan was fixed; it compares phases that differ in the evidence they contain, which is why we calibrate it with a constant-rate learner run on the same episodes. The size of the raw–summarized gap depends on the layout of the history: a terse one-line-per-trial record raised raw-history accuracy and lowered the use of coded and summarized histories, shrinking the gap to 1–3 points (Appendix J). Cover stories vary wording only and are not real-world datasets. In the Psych-101 comparison, human participants were told the hazard rate and had played earlier games, whereas the models' primary condition hides it, so that comparison is descriptive across conditions, and the human learning-rate fits are our re-analysis; the comparison's estimator is biased upward by about 0.09 and has power of about one half for differences as large as the human slope, so it is supporting evidence only and a null result is uninterpretable. The analysis plan was fixed before held-out evaluation by a timestamped commit, not registered with a third party; after that commit, and before any held-out model output existed, the hardware used for evaluation changed and the 72B model was dropped, both for compute reasons (Appendix K). Before that commit we had seen pilot results on an earlier version of the episodes, part of one model's control-condition results on development episodes, the rule-only baselines (no model) on held-out episodes, and the outputs of plumbing tests on a few held-out items, which were not analysed. We make no claim about neural mechanisms.

References

Adams, R. P.; and MacKay, D. J. C. 2007. Bayesian Online Changepoint Detection. *arXiv preprint arXiv:0710.3742*.

Ashkinaze, J.; Kurek, L.; Faisal, A.; Miao, T.; Joseph, M.; Budak, C.; and Gilbert, E. 2026. Information Discernment in Large Language Models. arXiv:2607.19355.

Bai, S.; Cai, Y.; Chen, R.; Chen, K.; Chen, X.; Cheng, Z.; et al. 2025. Qwen3-VL Technical Report. *arXiv preprint arXiv:2511.21631*.

Behrens, T. E. J.; Hunt, L. T.; Woolrich, M. W.; and Rushworth, M. F. S. 2008. Associative learning of social value. *Nature*, 456(7219): 245–249.

Behrens, T. E. J.; Woolrich, M. W.; Walton, M. E.; and Rushworth, M. F. S. 2007. Learning the value of information in an uncertain world. *Nature Neuroscience*, 10(9): 1214–1221.

Binz, M.; Akata, E.; Bethge, M.; Brändle, F.; Callaway, F.; Coda-Forno, J.; Dayan, P.; Demircan, C.; Eckstein, M. K.; Éltető, N.; Griffiths, T. L.; Haridi, S.; Jagadish, A. K.; Ji-An, L.; Kipnis, A.; Kumar, S.; Ludwig, T.; Mathony, M.; Mattar, M.; Modirshanechi, A.; Nath, S. S.; Peterson, J. C.; Rmus, M.; Russek, E. M.; Saanum, T.; Schubert, J. A.; Schulze Buschoff, L. M.; Singhi, N.; Sui, X.; Thalmann, M.; Theis, F. J.; Truong, V.; Udandarao, V.; Voudouris, K.; Wilson, R.; Witte, K.; Wu, S.; Wulff, D. U.; Xiong, H.; and Schulz, E. 2025. A foundation model to predict and capture human cognition. *Nature*, 644(8078): 1002–1009.

Binz, M.; and Schulz, E. 2023. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6): e2218523120.

Bonaccio, S.; and Dalal, R. S. 2006. Advice taking and decision-making: An integrative literature review, and implications for the organizational sciences. *Organizational Behavior and Human Decision Processes*, 101(2): 127–151.

Campbell, D.; Rane, S.; Giallanza, T.; De Sabbata, N.; Ghods, K.; Joshi, A.; Ku, A.; Frankland, S. M.; Griffiths, T. L.; Cohen, J. D.; and Webb, T. W. 2024. Understanding the Limits of Vision Language Models Through the Lens of the Binding Problem. In *Advances in Neural Information Processing Systems*, volume 37, 113436–113460.

Chen, Y. 2026. Trust Between AI Agents: Measuring Formation, Breakage, and Recovery, with Implications for Governing Multi-Agent Systems. arXiv:2606.14923.

Coda-Forno, J.; Binz, M.; Wang, J. X.; and Schulz, E. 2024. CogBench: a large language model walks into a psychology lab. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 9076–9108. PMLR.

Diaconescu, A. O.; Stecy, M.; Kasper, L.; Burke, C. J.; Nagy, Z.; Mathys, C.; and Tobler, P. N. 2020. Neural arbitration between social and individual learning systems. *eLife*, 9: e54051.

Feng, P.; Yang, S.; and Poria, S. 2026. Σ-Mem: An Online Reliability Memory for LLM-based Multi-Agent Systems. arXiv:2607.27958.

Fu, D.; Guo, R.; Khalighinejad, G.; Liu, O.; Dhingra, B.; Yogatama, D.; Jia, R.; and Neiswanger, W. 2024. IsoBench: Benchmarking Multimodal Foundation Models on Isomorphic Representations. In *First Conference on Language Modeling (COLM)*. ArXiv:2404.01266.

Ge, L.; Zhang, Y.; and Vorobeychik, Y. 2026. Mind the (DH) Gap! A Contrast in Risky Choices Between Reasoning and Conversational LLMs. *arXiv preprint arXiv:2602.15173*.

Gemma Team. 2025. Gemma 3 Technical Report. *arXiv preprint arXiv:2503.19786*.

Gupta, R.; Corona, R.; Ge, J.; Wang, E.; Klein, D.; Darrell, T.; and Chan, D. M. 2025. Enough Coin Flips Can Make LLMs Act Bayesian. *arXiv preprint arXiv:2503.04722*.

Hayes, W. M.; Yax, N.; and Palminteri, S. 2024. Large Language Models are Biased Reinforcement Learners. arXiv:2405.11422.

Hertwig, R.; Barron, G.; Weber, E. U.; and Erev, I. 2004. Decisions from Experience and the Effect of Rare Events in Risky Choice. *Psychological Science*, 15(8): 534–539.

Krishnamurthy, A.; Harris, K.; Foster, D. J.; Zhang, C.; and Slivkins, A. 2024. Can Large Language Models Explore In-Context? In *Advances in Neural Information Processing Systems*. ArXiv:2403.15371.

Lakens, D. 2017. Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science*, 8(4): 355–362.

Li, L.; Wang, Y.; Zhao, H.; Kong, S.; Teng, Y.; Li, C.; and Wang, Y. 2025. Reflection-Bench: Evaluating Epistemic Agency in Large Language Models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 36236–36264. PMLR.

Mathys, C.; Daunizeau, J.; Friston, K. J.; and Stephan, K. E. 2011. A Bayesian foundation for individual learning under uncertainty. *Frontiers in Human Neuroscience*, 5: 39.

Nassar, M. R.; Rumsey, K. M.; Wilson, R. C.; Parikh, K.; Heasly, B.; and Gold, J. I. 2012. Rational regulation of learning dynamics by pupil-linked arousal systems. *Nature Neuroscience*, 15(7): 1040–1046.

Nassar, M. R.; Wilson, R. C.; Heasly, B.; and Gold, J. I. 2010. An approximately Bayesian delta-rule model explains the dynamics of belief updating in a changing environment. *Journal of Neuroscience*, 30(37): 12366–12378.

Parkhi, O. M.; Vedaldi, A.; Zisserman, A.; and Jawahar, C. V. 2012. Cats and dogs. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 3498–3505.

Piray, P.; and Daw, N. D. 2021. A model for learning based on the joint estimation of stochasticity and volatility. *Nature Communications*, 12: 6587.

Piray, P.; and Daw, N. D. 2024. Computational processes of simultaneous learning of stochasticity and volatility in humans. *Nature Communications*, 15: 9073.

Schmied, T.; Bornschein, J.; Grau-Moya, J.; Wulfmeier, M.; and Pascanu, R. 2025. LLMs are Greedy Agents: Effects of RL Fine-tuning on Decision-Making Abilities. arXiv:2504.16078.

Schubert, J. A.; Jagadish, A. K.; Binz, M.; and Schulz, E. 2024. In-Context Learning Agents Are Asymmetric Belief Updaters. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 43928–43946. PMLR.

Soltani, A.; and Izquierdo, A. 2019. Adaptive learning under expected and unexpected uncertainty. *Nature Reviews Neuroscience*, 20(10): 635–644.

Tian, X.; Zou, S.; Yang, Z.; and Zhang, J. 2025. Identifying and Mitigating Position Bias of Multi-image Vision-Language Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 10599–10609.

Wang, H.; Ward, T. E.; and Zhang, L. 2026a. Comparative reversal learning reveals rigid adaptation in LLMs under non-stationary uncertainty. Accepted to IPMU 2026 (per arXiv comment), arXiv:2604.04182.

Wang, H.; Ward, T. E.; and Zhang, L. 2026b. Rigidity in LLM Bandits with Implications for Human-AI Dyads. arXiv:2603.07717.

Wang, W.; Gao, Z.; Gu, L.; Pu, H.; Cui, L.; Wei, X.; et al. 2025. InternVL3.5: Advancing Open-Source Multimodal Models in Versatility, Reasoning, and Efficiency. *arXiv preprint arXiv:2508.18265*.

Wulff, D. U.; Mergenthaler-Canseco, M.; and Hertwig, R. 2018. A meta-analytic review of two modes of learning and the description-experience gap. *Psychological Bulletin*, 144(2): 140–176.

Xiong, H.-D.; Ji-An, L.; Mattar, M. G.; and Wilson, R. C. 2023. Neural network modeling reveals diverse human exploration behaviors via state space analysis. In *Conference on Cognitive Computational Neuroscience (CCN)*.

Xue, J.; Wilson, R. C.; and Xiong, H. 2026. Counting on Thinking: Tracing Evidence Integration in Language Models. *arXiv preprint arXiv:2609.32932*.

Yaniv, I.; and Kleinberger, E. 2000. Advice Taking in Decision Making: Egocentric Discounting and Reputation Formation. *Organizational Behavior and Human Decision Processes*, 83(2): 260–281.

Zhang, J.; Yang, J.; and Wang, K. 2025. Large Language Models as Discounted Bayesian Filters. *arXiv preprint arXiv:2512.18489*.

Zhou, R.; Song, M.; Wu, X.; Cheng, S.; Yin, X.; Xie, Y.; Hao, Z.; Hua, W.; Pan, L.; Poria, S.; and Kan, M.-Y. 2026. Epistemic Context Learning: Building Trust the Right Way in LLM-Based Multi-Agent Systems. ICLR 2026 Workshop on Agentic AI in the Wild, arXiv:2601.21742.

Supplementary Material

A. Prompts and Cover Stories

Every prompt has four parts: instructions naming the two advisors and the cover story, the history in one of the formats of the main text, the current reports, and the question. Prompts are passed through each model's chat template as a single user message. For InternVL3.5 the template adds the model's default system message (in Chinese) and the suffix "Answer the question using a single word or phrase." All answer words are single tokens in every tokenizer used.

Table 3 lists the cover stories. The examples below are rendered by the experiment code from the frozen episode files; line breaks are as in the prompt and *[image]* marks where an image is inserted.

Example 1: raw history, weather, held-out episode, trial 6.

> You are observing a sequence of days. On every day, two weather forecasters, Tim and Joe, give conflicting forecasts for the day's weather (rain or dry). After each day, the actual weather is observed. A forecaster's accuracy may change over time.
> Past days:
> Day 1: Joe forecasts rain; Tim forecasts dry. Actual weather: dry.
> Day 2: Tim forecasts rain; Joe forecasts dry. Actual weather:

Cover story	Advisors	Answers	Report image
color	reporters	red/blue	color patch
weather	forecasters	rain/dry	cloud, sun
stock	analysts	up/down	trend arrow
sports	commentators	win/lose	happy/sad face
medical	doctors	positive/negative	plus/minus
news	news sources	true/false	tick, cross
pet (photo)	reporters	cat/dog	photograph

Table 3: Cover stories with the selected icon set (v3.1). Trials are called trials (color, pet), days, weeks, games, cases or stories. Icons are drawn by code and contain no answer words. Appendix G shows the original and selected medical and sports icons and identifies results obtained with the original set.

rain.
Day 3: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
Day 4: Joe forecasts rain; Tim forecasts dry. Actual weather: dry.
Day 5: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
Current day 6 (actual weather not yet observed):
Joe forecasts: dry
Tim forecasts: rain
Predict the actual weather on the current day. Answer with exactly one lowercase word: rain or dry.

The other formats change only the history lines. In the raw history with correctness, day 1 reads "Day 1: Joe forecasts rain; Tim forecasts dry. Actual weather: dry. (Joe was wrong; Tim was correct.)". In the correctness-coded history it reads "Day 1: Joe was wrong; Tim was correct.". In the summarized history the five history lines and the "Past days:" line are replaced by "Summary of all 5 past days: Joe was correct 0 times; Tim was correct 5 times.". In the correctness chart they are replaced by "Past days 1-5 are summarized in this chart (days run left to right in four lines of 20; for each day, a green square means that forecaster was correct, a gray square means that forecaster was wrong, an empty square means not yet played):" followed by the chart (448 px; one row per advisor in each line). In the judgment readout the current reports and question are replaced by "Before day 6: which forecaster is currently more likely to be correct? Answer with exactly one name: Joe or Tim.".

Example 2: stated reliability, weather, development episode.

You are observing a sequence of days. On every day, two weather forecasters, Jim and Ken, give conflicting forecasts for the day's weather (rain or dry). After each day, the actual weather is observed. A forecaster's accuracy may change over time.
Summary of past performance: Jim was correct on 80% of past days; Ken was correct on 20% of past days.
Current day (actual weather not yet observed):
Ken forecasts: rain
Jim forecasts: dry
Predict the actual weather on the current day. Answer with exactly one lowercase word: rain or dry.

The stated-counts control replaces the summary line with "Summary of the last 10 days: Jim was correct 8 times; Ken was correct 2 times.".

Example 3: photo cover story with image reports, held-out episode, trial 4.

You are observing a sequence of trials. On every trial, two reporters, Jon and Jim, give conflicting reports about a hidden animal. After each trial, the true animal is revealed. A reporter's accuracy may change over time.
Past trials:
Trial 1: Jon reports cat; Jim reports dog. True animal: cat.
Trial 2: Jon reports cat; Jim reports dog. True animal: cat.
Trial 3: Jim reports dog; Jon reports cat. True animal: cat.
Current trial 4 (true animal not yet revealed):
Jim reports: *[image]*
Jon reports: *[image]*
Predict the true animal on the current trial. Answer with exactly one lowercase word: cat or dog.

With text reports the images are replaced by the answer word.

Binding variants. The binding controls change only how the current reports of Example 3 are attached to the advisors (stated reliability, image reports). *Name on image*: the lines are as in Example 3, but each image carries its advisor's name printed in a banner above the picture. *Composite image*: the two report lines are replaced by "Reports (one panel per reporter, name above each panel): *[image]*", one image with two panels, each headed by an advisor's name. *Image after name*: each report is given as "*[image]* (the image above was reported by Jim)", so the attribution follows the image. The composite-read check shows only the composite image and asks "The image has two panels, each with a name above it. What animal is in Jim's panel?".

B. Episode Generation and Counterbalancing

Schedules. Each episode has 80 trials in eight blocks of 10. Advisor A is the advisor favored on trial 1. Stable-first episodes have a stable phase (A correct with probability 0.8) followed by a volatile phase with blocks at 0.2, 0.8, 0.2, 0.8; volatile-first episodes have a volatile phase with blocks at 0.8, 0.2, 0.8, 0.2 followed by a stable phase at 0.8. Every episode therefore has four reversals (trials 41, 51, 61, 71 in stable-first and 11, 21, 31, 41 in volatile-first episodes). Advisor B is always correct when A is wrong. For comparison, the advice schedule of Behrens et al. (2008) (Supplementary Information) has 120 trials: 75% correct on trials 1–30, alternation between 80% correct and 80% incorrect every 10 trials on trials 31–70, and 85% incorrect on trials 71–120. Our volatile phase matches their alternation; our stable phase does not.

Stratification. Within each block, A is correct on exactly $10p$ trials (8 or 2) in random order, and each outcome label is the truth on exactly five trials. This keeps realized accuracy equal to the schedule in every episode. It also fixes outcome quotas, which allow a shortcut that ignores the advisors: predicting the label with more occurrences remaining in the current block's quota (which presupposes knowing the quota) reaches 0.661 accuracy on the development episodes. We therefore also use a robustness set (below).

Names, order and counterbalancing. Advisor names are drawn from 16 common English first names that are single tokens in all tokenizers. The order in which the two reports are listed is random on each trial. Phase order alternates across episodes, the order in which the instructions introduce the two names alternates in pairs, and text cover stories rotate in blocks of four episodes, so cover story, phase order and introduction order are fully crossed. Each episode has a counterbalanced copy with the advisor names swapped (A keeps its role), the answer-label mapping reversed and the introduction order reversed; outcomes and images are identical. Analyses average the label-aligned logits within each pair.

Splits and seeds. The main set has 16 development episodes (color, weather, stock and sports cover stories) and 48 held-out episodes (eight per text cover story), 128 with counterbalanced copies, generated with seed 20261111. Pet photos come from the Oxford-IIIT Pet dataset (CC BY-SA 4.0): 128 photos from the official trainval split for development episodes and 256 from the test split for held-out episodes, with no overlap. The robustness set has 24 held-out episodes (48 with copies; seed 20261112) in which correctness is drawn independently on each trial, outcome labels are drawn independently, and volatile-phase dwell times are drawn uniformly from 6–16 trials. All data are regenerated from seeds and checked against stored SHA-256 hashes.

C. Control Conditions and Decomposition: Full Results

Table 5 gives each control condition with text and image reports, and Table 6 the stepwise differences between history formats on the main task. Image reports lowered accuracy in almost every control condition, including stated reliability (.65–.84 instead of .92–1.00); the one exception was Gemma-3-27B with a short raw history (.612 vs. .603). The image-report results use the selected icons (v3.1, Appendix G), which every model named correctly on every single-image item (photos: .986–1.00), so these drops reflect the use of image reports rather than recognition of a single image. **[DRAFT: development episodes (16) and four models; replace with held-out results]**

	Stated		Short		Long		Single
Model	%	counts	raw	coded	raw	coded	trial
Qwen3-VL-8B	.917	.998	.701	1.00	.533	.681	1.00
InternVL3.5-8B	1.00	1.00	.638	.996	.562	.676	1.00
Gemma-3-27B	.980	.973	.603	1.00	.540	.850	1.00
Qwen3-VL-32B	1.00	1.00	.607	1.00	.545	.750	1.00
InternVL3.5-38B	**??**	**??**	**??**	**??**	**??**	**??**	**??**

Table 4: Control conditions on development episodes, text reports (accuracy on control items, averaged over counterbalanced copies; image reports in Appendix C). Stated: reliability given as a percentage or as counts. Short/long: an easy history in which one advisor is clearly better, given raw or correctness-coded. Single trial: who was correct on one shown trial. Full results by cover story in Appendix C.

D. Normative Simulation

We ran two simulations without any VLM to check the design. Both score a rule by the proportion of trials ($t \geq 2$) on which the advisor it favours is correct; ties score 0.5.

Design simulation. The first simulation draws outcomes independently from the schedule (300 episodes per setting) and compares BOCPD (posterior predictive, $h \in \{1/8, 1/15, 1/30\}$), majority counts over the last $K \in \{3, 5, 7, 10, 15, 20, 30\}$ trials, counts over all past trials, and counts since the last true reversal. Across 20 settings (accuracy 0.65–0.85; phase lengths 40/40, 60/60 and 80/40; dwell 10 or 15 trials), the best BOCPD minus the best fixed window ranged from -0.004 to $+0.008$. For our setting (0.8; 40/40; dwell 10), accuracy was 0.687 for the best BOCPD ($h = 1/8$), 0.683 for $h = 1/15$, 0.690 for the best window ($K = 3$), 0.632 for counts over all trials and 0.739 for counts since the last reversal. Conclusions were the same for accuracies from 0.65 to 0.85, and accuracy varied by about one percentage point for hazard rates between 1/8 and 1/30.

Rules given to the VLM. The second simulation applies exactly the rules used to build the summarized histories (Appendix E) to the actual episodes. Table 7 gives the results. On held-out episodes, the 5-trial window (.692 [.685, .698]) is about 1.4 percentage points above the change-point summary (.678 [.666, .688]); counts over all trials (.632) are 6.0 points below the 5-trial window and 4.6 points below the change-point summary. On the robustness set, the 5-trial window (.675) again leads the change-point summary (.663) and the reversal-aligned reference is highest (.735), but the remaining rules are ordered differently: on held-out episodes counts over all trials (.632) exceed the 10-trial (.612) and 20-trial (.579) windows, whereas on the robustness set the 10-trial (.635) and 20-trial (.604) windows exceed counts over all trials (.594). These differences hold for the tested settings only; we do not claim that online rules are equivalent in general. The reversal-aligned summary uses only reversals up to $t-1$ and so learns of a reversal one trial late; it is a reference, not an upper bound on any strategy. Likewise, 0.8 is the accuracy of an observer who knows the current accuracy, not a strict ceiling (Appendix B).

These rule baselines were computed on held-out episodes before any model results on the main data were seen. They are properties of the episode generator and involve no model behavior.

E. Change-Point Controller and Summary Variants

Change-point detection. Let $o_t = 1$ if advisor A was correct on trial t. Before trial t, the controller runs Bayesian online change-point detection (Adams and MacKay 2007) on $o_1, \ldots, o_{t-1}$ with a Bernoulli likelihood, a Beta(1,1) prior for each run and a constant hazard rate $h = 1/15$. This rate lies between the reversal rates of the volatile (1/10) and stable phases and was not tuned (Appendix D). The controller takes the maximum a posteriori run length r, sets it to at least 3, and summarizes trials $\max(1, t-r)$ to $t-1$.

Summary variants. All variants use the same sentence as the summarized history, restricted to a range of trials

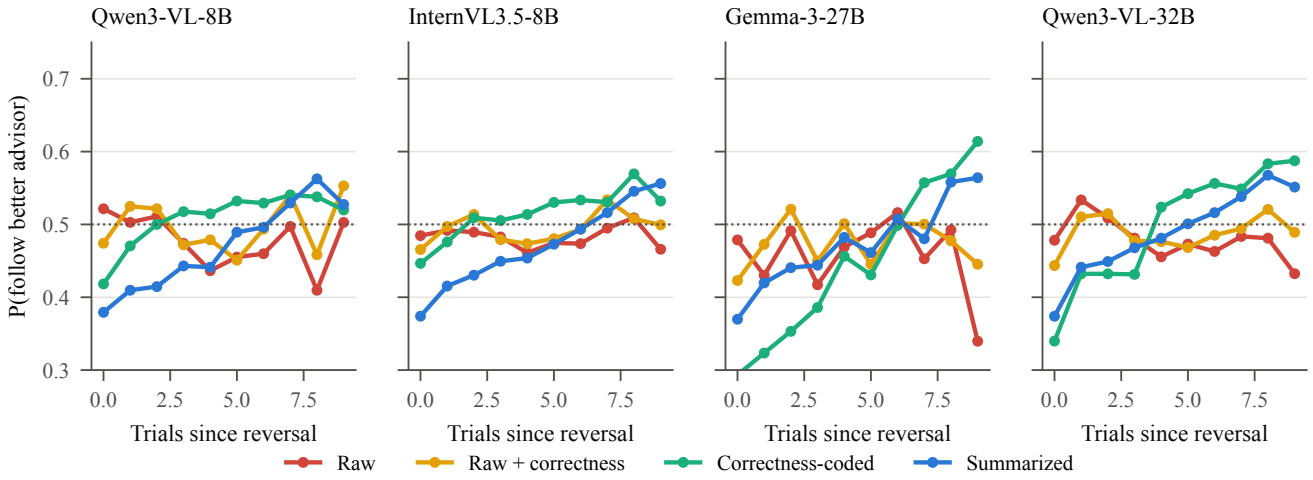


Figure 5: Probability of following the currently better advisor as a function of trials since a reversal, by history format. **[DRAFT: development episodes (16) and four models; replace with held-out results] [TODO: add window and change-point summaries]**

("Summary of the last n days (days s–$(t-1)$): Ken was correct x times; Sam was correct y times."). The variants are: all past trials (the primary summary); the last $K \in \{5, 10, 20\}$ trials; the run since the last inferred change-point (BOCPD); and the trials since the last true reversal up to $t-1$ (reversal-aligned reference). Two further conditions truncate the raw history instead of summarizing it: to the BOCPD run, or to the last 10 trials. The window length used in the secondary tests is, for each model, the one with the highest accuracy on development episodes, fixed before held-out evaluation; Table 9 reports the selected length and all variants, and Table 10 compares the model with the rule alone.

What the controller is. The controller is a training-free rule outside the model; it sees only the outcome sequence and does not read model states. It is motivated by change-point models of human learning (Nassar et al. 2010, 2012), but we use it as an integration rule, not as a model of the VLM or of the brain.

F. Computational Models and Parameter Recovery

Models. Let $o_t = 1$ if advisor A was correct on trial t and y_t the model's preference for A. Each computational model maps $o_1, \ldots, o_{t-1}$ to a belief q_t that A is correct and predicts $P_t = \sigma(\beta(q_t - 0.5) + b)$, with inverse temperature β and bias b. M0 is a constant bias ($P_t = \sigma(b)$). Win-stay/lose-shift follows the advisor who was correct on the previous trial ($P_t = \sigma(b + w(2o_{t-1} - 1))$). Count models use $q_t = (s+1)/(n+2)$ over all past trials or over the last k trials. M1 is a delta rule, $q_{t+1} = q_t + \alpha(o_t - q_t)$ with $q_1 = 0.5$; M2 uses separate learning rates in the stable and volatile phases; M3 uses separate rates for outcomes that confirm or disconfirm the currently trusted advisor. M4 comprises a Bayesian change-point observer (hazard rate from a grid, or marginalized over it) and a volatility-tracking learner after Behrens et al. (2007), each either with a fitted readout or reading out its predictive probability directly.

Fitting and comparison. Because y_t is a preference rather than a sampled choice, models are fitted by minimizing the soft cross-entropy $-\sum_t [y_t \log P_t + (1 - y_t) \log(1 - P_t)]$, a pseudo-likelihood. Models are compared by leave-one-episode-out cross-validation: parameters are fitted jointly on all other episodes of a model and history format and scored on the left-out episode. Information criteria are reported only as heuristics. Following the analysis plan, learning-rate analyses are run only for conditions in which a model can use reliability at all (stated reliability ≥ 0.8 in the control conditions) and tracks the better advisor in the stable phase (probability ≥ 0.6); on development episodes, stable-phase following under raw histories was .53–.57 in every model, below this criterion, whereas it was .86–.98 with correctness-coded and summarized histories.

Effective learning rate. The update of the model's preference in logit space after each outcome, regressed on the signed outcome separately for the stable and volatile phases, measures how strongly the model responds to new outcomes. This statistic is not zero for a learner with a single constant learning rate: preferences are more extreme in the stable phase, where the logit stretches updates. We therefore calibrate it against a null in which M1, fitted to the same episode, generates surrogate preferences with the same residual noise (200 surrogates); the calibrated volatile-minus-stable difference and its z score against the null indicate whether a model updates more in the volatile phase than a constant-rate learner would.

Recovery. In simulations with M1–M3 (100 synthetic learners each, logit noise s.d. 0.3), learning rates were recovered with correlations of .998 (M1; M2 stable and volatile rates) and .89–.99 (M3). Model identification was asymmet-

Condition	Reports	Qwen3-8B	IVL-8B	Qwen3-32B	Gemma-27B	IVL-38B
Stated %	Text	.917	1.00	1.00	.980	**??**
	Image	.835	.806	.656†	.647	**??**
Stated counts	Text	.998	1.00	1.00	.973	**??**
	Image	.855	.779	.644†	.647	**??**
Short raw	Text	.701	.638	.607	.603	**??**
	Image	.629	.545	.504†	.612	**??**
Short coded	Text	1.00	.996	1.00	1.00	**??**
	Image	.940	.746	.683†	.683	**??**
Long raw	Text	.533	.562	.545	.540	**??**
	Image	.540	.518	.409†	.507	**??**
Long coded	Text	.681	.676	.750	.850	**??**
	Image	.598	.429	.477†	.603	**??**
Single trial	Text	1.00	1.00	1.00	1.00	**??**
Stated %, name on image	Image	.821	.830	.759	.734	**??**
Stated %, composite image	Image	.748	.842	.770	.824	**??**
Stated %, image after name	Image	.596	.696	.549	.520	**??**
Composite read	Image	1.00	1.00	1.00	1.00	**??**
Chart read	Chart	.630	.688	.672	.604	.714

Table 5: Control conditions on development episodes (accuracy on control items, averaged over counterbalanced copies and pooled over cover stories). Lower block: binding variants of the stated-reliability condition with image reports (advisor name printed on the image; both reports in one composite image; image given after the name), reading one panel of the composite image, and reading one cell of the correctness chart. Qwen3: Qwen3-VL; IVL: InternVL3.5. Image reports use the selected icons (v3.1). †Partial run (413 of 448 items per cell). Results by cover story are in the released analysis output.

Model	Raw+corr.	Coded	Chart	Summary−coded	Summary	Position
Qwen3-VL-8B	+.049	+.155	–	−.026	+.129	1.20 [1.08, 1.37]
InternVL3.5-8B	+.062	+.168	–	−.011	+.157	1.07 [1.02, 1.13]
Gemma-3-27B	+.035	+.172	–	+.018	+.190	0.91 [0.85, 0.96]
Qwen3-VL-32B	+.065	+.164	–	+.005	+.169	0.97 [0.85, 1.08]
InternVL3.5-38B	**??**	**??**	**??**	**??**	**??**	**??**

Table 6: **[DRAFT: development episodes (16) and four models; replace with held-out results]** Decomposition of the raw–summarized gap: paired differences in main-task accuracy relative to raw history (except Summary−coded) with 95% bootstrap CIs (format: mean [low, high]). Position: (coded − raw)/(summary − raw), reported only when the summary–raw gap exceeds the prespecified threshold and its bootstrap CI excludes 0; otherwise "n/i".

Summary rule	Development	Held-out	Robustness
All trials	.630	.632	.594
Last 5 trials	.698	.692	.675
Last 10 trials	.620	.612	.635
Last 20 trials	.578	.579	.604
Change-point (BOCPD)	.678	.678	.663
Since last reversal	.739	.744	.735

Table 7: Accuracy of the summary rules alone (no model) on development (16), held-out (48) and robustness (24) episodes. 95% bootstrap CIs over episodes (2,000 resamples), held-out: last 5 trials [.685, .698]; change-point [.666, .688]; all trials [.628, .636].

Baseline (no model)	Held-out	Robustness
Always follow A	**??**	**??**
Known current accuracy	**??**	**??**
BOCPD posterior	**??**	**??**
Outcome quota (knows block quota)	**??**	**??**

Table 8: Additional model-free baselines on the trial mask used for the model runs (95% CIs in the analysis output).

ric: data from M1 were always classified as M1, but data from M2 were classified as M2 in 40% of cases (70% when the two rates differed by more than 0.2) and data from M3 rarely as M3 (10%). The sign of the M2 rate difference was recovered in every case with a clear difference. Phase-specific rates are thus estimable, whereas model comparison favors the simpler M1 unless the difference is large; we therefore base conclusions about volatility on the calibrated effective learning rate rather than on model selection alone. Table 13 checks the strategy and kernel analyses of the main text on synthetic agents run on the same episodes.

		Summarized history						Truncated raw history	
Model	Selected K	All trials	Last 5	Last 10	Last 20	Change-point	Since reversal	Change-point	Last 10
Qwen3-VL-8B	??	??	??	??	??	??	??	??	??
InternVL3.5-8B	??	??	??	??	??	??	??	??	??
Gemma-3-27B	??	??	??	??	??	??	??	??	??
Qwen3-VL-32B	??	??	??	??	??	??	??	??	??
InternVL3.5-38B	??	??	??	??	??	??	??	??	??
Rule alone	–	.632	.692	.612	.579	.678	.744	–	–

Table 9: Main-task accuracy for every summary variant and the truncated raw histories (held-out). Rule alone: the summary's majority rule without a model (Table 7).

Model	Summ.	VLM	Rule	Δ [90% CI]	Agree
Qwen3-VL-8B	All	.616	.630	−.014 [−.026, −.004]	.926
	CP	??	??	??	??
	Selected	??	??	??	??
InternVL3.5-8B	All	.628	.630	−.002 [−.007, .004]	.975
	CP	??	??	??	??
	Selected	??	??	??	??
Gemma-3-27B	All	.625	.630	−.004 [−.011, .003]	.946
	CP	??	??	??	??
	Selected	??	??	??	??
Qwen3-VL-32B	All	.632	.630	+.002 [−.002, .007]	.994
	CP	??	??	??	??
	Selected	??	??	??	??
InternVL3.5-38B	All	??	??	??	??
	CP	??	??	??	??
	Selected	??	??	??	??

Table 10: **[DRAFT: development episodes (16) and four models; replace with held-out results]** VLM with summarized history vs. the summary's rule alone on identical trials. Agree: proportion of non-tied trials on which the model's choice equals the rule's. Equal accuracy does not imply the same strategy; agreement is reported for that reason.

G. Icon Design, Multimodal Binding and Chart Legibility

The legibility and binding controls use development episodes only and are exploratory; the comparison of text and image reports on the main task uses held-out episodes. All use the selected icons (v3.1) described below. Every model named every single report image correctly (InternVL3.5-8B: .986 for pet photos), so the remaining costs are not recognition failures (Table 14). With stated reliability, accuracy was lowest when the image followed the advisor's name (.52–.70) and higher when the name was printed on the image (.73–.83) or both reports were shown in one composite image (.75–.84), although each panel of a composite image was read correctly on every item. With correctness-coded histories, image reports lowered main-task accuracy from .608–.642 to .529–.553. Reading a single cell of the correctness chart was below the 0.8 gate for every model (.60–.71).

Icon design and examples. Icons replace the answer word with a picture in the image-report conditions. Figure 7 shows the actual deterministic renderings used in the icon probe, at 224 pixels; InternVL3.5 receives 448-pixel renderings from the same drawing functions. In the original medical pair, two red test-strip lines indicate positive and one indicates negative. The selected pair uses a dark plus or minus inside the same circular outline. In sports, the original gold trophy and red cross on a grey ball encode win and lose; the selected pair uses a yellow happy face and a grey-blue sad face. Labels and selection markers in the figure are reader annotations and are absent from the report images. These are synthetic symbols for the answer labels, not clinical scans or photographs of game outcomes. All other cover-story icons are unchanged.

Development-only selection. We tested five medical pairs and four sports pairs with Qwen3-VL-8B and InternVL3.5-8B before held-out evaluation. The single-image probe asks for each of the two labels under both answer orders (four prompts per pair and model). We required all four responses to be correct for both models, then selected the eligible pair with the highest mean accuracy over four image-report controls and both models (Table 15). The controls use stated reliability, a short raw history, a short correctness-coded history, and a long correctness-coded history. Each downstream cell contains 64 items after averaging label-aligned logits across the counterbalanced episode copies. This selected plus/minus and happy/sad faces. Enlarging the test strip left naming accuracy at .50 for both models, so size alone did not resolve that failure. Coloured plus/minus signs scored higher downstream but failed the InternVL3.5 naming criterion.

Model	History	Copy last outcome	Follow last correct	Last 5	All-trial counts	BOCPD	Label prior
Qwen3-VL-8B	Raw	0.24	−0.02	−0.06	0.01	−0.05	−0.20
	Summary	0.02	0.07	0.13	0.18	0.15	0.03
InternVL3.5-8B	Raw	0.27	0.02	0.01	0.01	0.03	0.01
	Summary	0.00	0.08	0.15	0.39	0.17	0.01
Gemma-3-27B	Raw	0.35	−0.04	−0.06	0.02	−0.06	−0.15
	Summary	0.00	0.07	0.15	0.25	0.15	0.02
Qwen3-VL-32B	Raw	0.28	−0.01	−0.03	0.01	−0.02	−0.16
	Summary	0.00	0.07	0.13	0.79	0.13	0.01
InternVL3.5-38B	Raw	??	??	??	??	??	??
	Summary	??	??	??	??	??	??

Table 11: **[DRAFT: development episodes (16) and four models; replace with held-out results]** Agreement (Cohen's κ) between the model's choices and simple strategies. κ corrects for chance agreement, so a rule that almost always picks one advisor can have high raw agreement and low κ. Copy last outcome: predict the previous trial's outcome label. Follow last correct: follow the advisor who was correct on the previous trial. Label prior: always predict the same label.

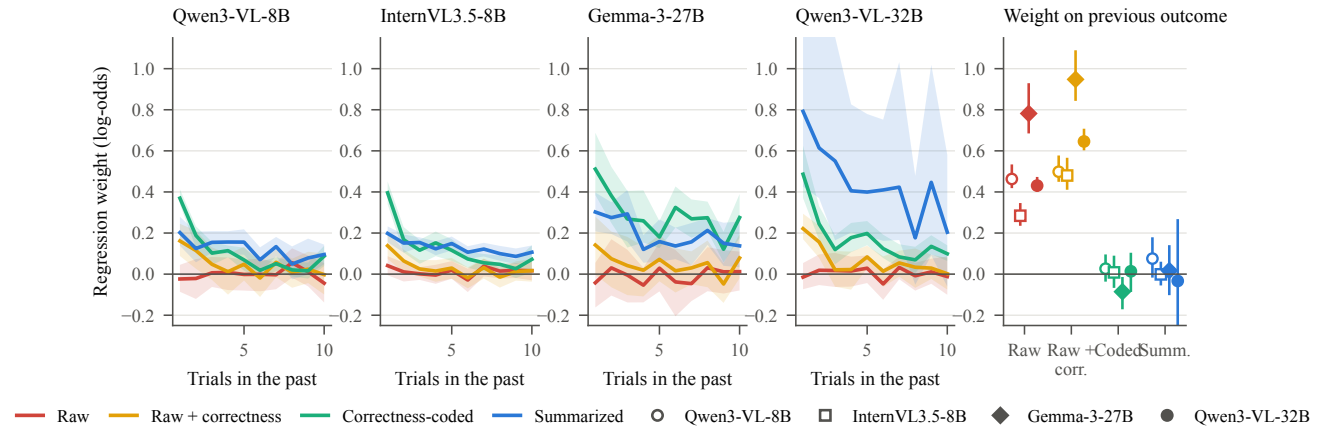


Figure 6: Integration kernels per model (shaded: 95% bootstrap CI over episodes) and, right, the weight on repeating the previous outcome by history format (bars: 95% CIs). **[DRAFT: development episodes (16) and four models; replace with held-out results]**

Results and limits. The downstream mean increases from .521 to .668 for medical and from .588 to .678 for sports. Both selected pairs reach 4/4 in the naming probe for both models. However, stated-reliability accuracy pooled over these two image-report cover stories is .773 for Qwen3-VL-8B and .727 for InternVL3.5-8B, below the .80 gate. Single-image recognition therefore does not establish successful use of image reports in the advice-taking task. Selection and evaluation use the same development controls, so these gains are descriptive and do not estimate held-out improvement.

Versioning and main-task check. This development-driven change is deviation D1 from the frozen plan. The selected icons define v3.1; v3 remains reproducible and original outputs are retained separately. Affected image-report and binding jobs require reruns. Text reports, pet photographs, the correctness chart, and the primary text-report comparison are unchanged. On the main task with correctness-coded history and image reports (16 development episodes, both counterbalanced copies, trials $t \geq 2$), the selected icons raised mean accuracy over the original ones for both models: Qwen3-VL-8B from .51 to .62 (medical) and from .56 to .67 (sports), and InternVL3.5-8B from .50 to .55 and from .52 to .55 (paired differences over 32 episode copies: +.11 [.06, .16], +.11 [.09, .14], +.05 [.03, .06], +.03 [.02, .05]; 95% bootstrap CIs). Accuracy remained below the same condition with text reports (Qwen3-VL-8B .57 and .70; InternVL3.5-8B .79 and .77), so image reports still add a cost beyond outcome evaluation, now with icons that both models name correctly. The selected icons also generalized beyond the two models used to choose them: Qwen3-VL-32B, InternVL3.5-38B and Gemma-3-27B named them correctly on every item. Tables 5 and 14 report the rerun with the selected icons.

H. Reasoning Mode

Reasoning is tested on development episodes only, with each episode's own text cover story and text reports. Qwen3-VL-8B-Thinking, the reasoning checkpoint of the same model family and size as Qwen3-VL-8B-Instruct, answers with its native reasoning template; the direct answers of Qwen3-VL-8B-Instruct on the same items are the reference. Because the

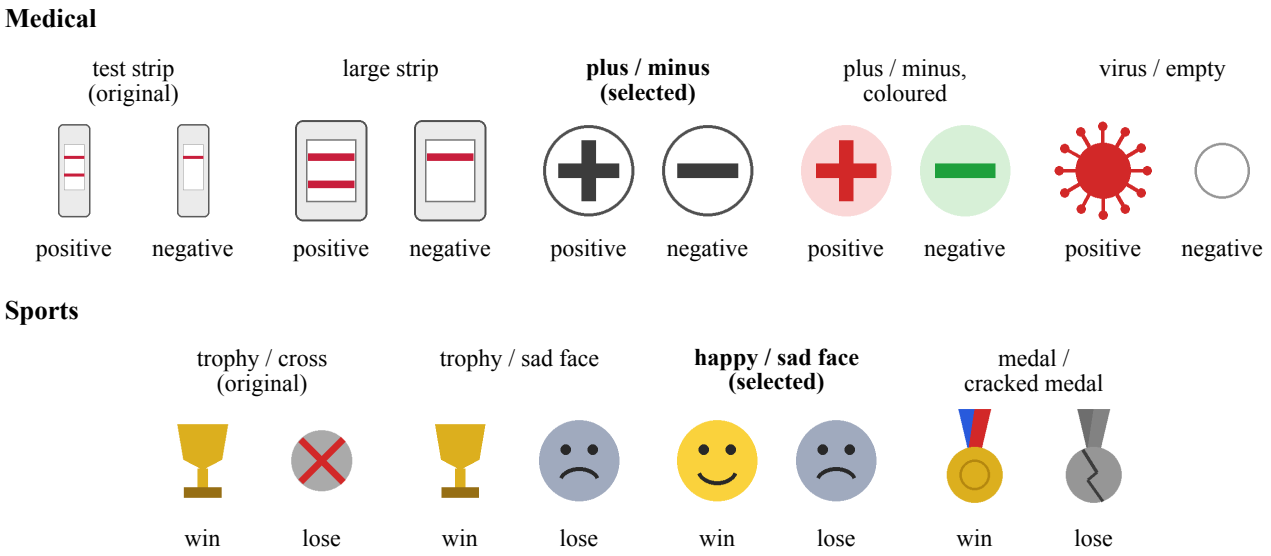


Figure 7: Actual candidate report icons for medical (positive/negative) and sports (win/lose). Original marks the frozen v3 pairs; selected marks their v3.1 replacements. Labels and headings are shown for the reader and are not embedded in model inputs. All candidates, including those not selected, are evaluated in Table 15.

Model	History	Best	NLL/trial	Vol.−stable	z
Qwen3-VL-8B	Raw	??	??	??	??
	Summary	??	??	??	??
InternVL3.5-8B	Raw	??	??	??	??
	Summary	??	??	??	??
Gemma-3-27B	Raw	??	??	??	??
	Summary	??	??	??	??
Qwen3-VL-32B	Raw	??	??	??	??
	Summary	??	??	??	??
InternVL3.5-38B	Raw	??	??	??	??
	Summary	??	??	??	??

Table 12: Computational-model fits on held-out episodes: best model by leave-one-episode-out cross-validated soft cross-entropy per trial, and the calibrated volatile-minus-stable difference in effective learning rate with its z score against the constant-rate null.

two are different checkpoints, the comparison bounds what reasoning adds at this scale but does not isolate reasoning within one checkpoint; for compute reasons we did not run larger reasoning models or a step-by-step prompt. The subset comprises the main task at trials 20, 40, 60 and 80 with raw and summarized history, and the stated-reliability, short and long histories and their correctness-coded versions at trials 40 and 80. Generation is greedy with up to 16,384 new tokens. We report free-answer accuracy, the valid-answer rate, truncation and forced-answer rates, and token cost separately; a probability read after the generated reasoning is conditional on that reasoning and is not compared directly with the direct-answer readout. Table 16 gives the results. **[TODO: one paragraph after results]**

I. Psych-101 Protocol and Additional Fits

Task and data. In the changing bandit of Xiong et al. (2023), participants repeatedly choose between two slot machines whose payoffs stay constant until a change point, which occurs with a hazard rate h; payoffs are noise-free, so a normative learner forgets the value of the unchosen machine at rate $\phi = h$. We use the 30 participants in Psych-101 (Binz et al. 2025) and their three 100-trial games ($h = 0.1, 0.2, 0.4$; 90 games, 9,000 trials). Participants were told h, and each 100-trial game directly preceded a 1,500-trial game with the same h, so by their second and third 100-trial games participants had played 1,600 and 3,200 trials; models see each game on its own.

Replay. Models are conditioned on each participant's previously executed behavior (Binz et al. 2025): before trial t the model sees the original instructions and the participant's choices and payoffs on trials $1, \ldots, t-1$, and is asked which machine to press (one letter). The readout is the preference between the two letters from the first-token logits, as in the main task. In the primary condition the hazard rate is removed from the instructions (hidden); in a secondary condition the original instructions state it (told). Model and participant thus share the within-game history but not the hazard information (hidden) or the prior experience (both conditions).

Fits. We fit the myopic model-free learner of Xiong et al. (2023), which updates the chosen machine's value and forgets the unchosen one toward the midpoint at rate ϕ, with a perseveration term, separately for each hazard rate. Humans give binary choices, so for comparability we draw 20 binary choice sequences from the model's preferences in each game, fit each, and take the median. The primary statis-

Synthetic agent	Best strategy	κ	Copy w	Lag-1 w
Copy last outcome ($w = 3$)	Copy last outcome	1.00	3.00	0.00
Delta rule ($\alpha = 0.3$, $\beta = 8$)	BOCPD	0.94	0.00	1.20
50% copy, 50% delta	Copy last outcome	0.49	1.21	0.56
Last-5 counts	Last 5	1.00	0.02	1.08
All-trial counts	All-trial counts	1.00	0.05	0.27
Label prior ($w = 3$)	Label prior	1.00	0.00	0.00

Table 13: Strategy and kernel analyses applied to synthetic agents on the main episodes (development and held-out; no VLM). Best strategy: highest Cohen's κ; Copy w and Lag-1 w: kernel regression weights on repeating the previous outcome and on the advisor correct one trial back. For the delta rule the fitted kernel decays geometrically with ratio 0.700 (true $1 - \alpha = 0.7$); the last-5 agent has a flat kernel up to lag 5. Strategies are correlated on this schedule, so the delta rule agrees best with BOCPD; the kernel, not the best-matching strategy, identifies it. Results on the robustness set were the same.

	Legibility controls			Main-task accuracy, correctness-coded history		
Model	Image naming	Composite read	Chart read	Text reports	Image reports	Chart history
Qwen3-VL-8B	1.000	1.000	.630	.642	.549	–
InternVL3.5-8B	.988	1.000	.688	.639	.553	–
Gemma-3-27B	1.000	1.000	.604	.608	.534	–
Qwen3-VL-32B	1.000	1.000	.672	.627	.529	–
InternVL3.5-38B	1.000	**??**	.714	**??**	**??**	**??**

Table 14: Multimodal reports (legibility controls on development episodes, exploratory; selected icons, v3.1). Image naming: single-image naming over all cover stories (photos and icons). Main-task columns: held-out episodes with correctness-coded history, which removes outcome evaluation; because every model names the images, the drop with image reports reflects their use in context, including advisor binding. Chart history: not yet run. Chart read below the prespecified 0.8 gate means chart-history results for that model are not interpreted.

tic is each participant's slope of ϕ on h; the comparison is the paired difference between the model's and the human's slope (Wilcoxon test, participant-bootstrap 95% CI), reported next to the model told versus human comparison and the model's told minus hidden contrast. The human fits are our re-analysis; Xiong et al. (2023) report no learning rates by hazard rate. In our re-analysis, the median human ϕ rose from .079 ($h = 0.1$) to .110 and .255 ($h = 0.4$; per-participant slope 0.47 per unit h, positive in 73% of participants, Wilcoxon $p = .008$), adjusting for game position did not remove the increase (0.64 [0.23, 1.05]), and in the following 1,500-trial games the increase was weaker (median ϕ .076, .114, .131; slope test $p = .029$).

Calibration of the human–model comparison. We simulated the complete pipeline 200 times per condition. Simulated participants follow each real participant's fitted strategy at each hazard rate and play on the real value sequences; the simulated model follows the same strategy with its forgetting-rate slope shifted by δ and is replayed on the simulated participant's history; both are then fitted exactly as the real data (20 sampled choice sequences per model episode, median fit, paired Wilcoxon test and bootstrap CI). With $\delta = 0$, the test rejected at $p < .05$ in 6.0% of replications (Monte Carlo SE 1.7%) and the 95% CI covered the true difference in 93.5%, but the estimated difference was biased upward by about $+0.09$. Power was 0.49 for $\delta = -0.5$ and 0.59 for $\delta = +0.5$, where CI coverage was 0.87 and 0.96. We therefore treat the comparison as supporting and descriptive (main text).

Image and chart formats. With the hazard rate hidden, the changing bandit is also run with the last five payoffs shown as pictures of the machine's display instead of text (224 px), and with the whole history drawn as one scatter chart of payoff by trial for each machine (448 px). Each has a perception check: reading the number on one machine picture (four options; 450 items), and reading which machine was pressed, or which gave more points, at a given trial of the chart (540 items). We report each format's fits next to its perception check, since a model that cannot read the pictures or the chart cannot be expected to learn from them. **[TODO: results for image and chart formats and their perception checks]**

J. Robustness

[TODO: one paragraph after results]

Which part of the raw history blocks the correctness statement. Appending who was correct to the raw history recovered only part of the summary–raw gap, and the weight on repeating the previous outcome stayed large. To find out why, we kept the layout and length of that history and replaced parts of each past trial with a dash ("Day 3: Ken forecasts —; Sam forecasts —. Actual weather: rain. (Ken was correct; Sam was wrong.)"): the outcome label, the two reports, or both. With both replaced the history carries the same information as the correctness-coded history at nearly the length of the raw history with correctness. On development episodes (Table 20), replacing the outcome label raised

Icon pair	Naming probe	Stated %	Short raw	Short coded	Long coded	Mean
Medical (positive / negative)						
Test strip (original)	.50/.50	.61/.36	.58/.62	.62/.36	.56/.45	.52
Large test strip	.50/.50	.61/.53	.58/.59	.73/.77	.62/.56	.62
Plus / minus (selected)	1.00/1.00	.70/.67	.84/.62	1.00/.50	.61/.39	.67
Plus / minus, coloured	1.00/.50	.72/.70	.77/.64	1.00/.95	.61/.52	.74
Virus / empty circle	.75/.75	.61/.92	.59/.59	1.00/.41	.48/.39	.62
Sports (win / lose)						
Trophy / cross (original)	1.00/.50	.61/.64	.52/.55	.72/.62	.61/.44	.59
Trophy / sad face	1.00/1.00	.66/.73	.55/.48	.84/.83	.61/.44	.64
Happy / sad face (selected)	1.00/1.00	.84/.78	.50/.52	.80/1.00	.50/.48	.68
Medal / cracked medal	1.00/.75	.61/.64	.53/.41	.64/.39	.48/.50	.53

Table 15: Icon comparison on development items, image reports (Qwen3-VL-8B / InternVL3.5-8B). Naming probe: accuracy over two labels $\times$ two answer orders (four prompts per model and icon pair). Downstream columns: hard accuracy over 64 items per cell after averaging counterbalanced-copy logits; Stated % denotes stated reliability. Mean averages the four downstream conditions and both models, excluding the naming probe. Selection requires 4/4 naming accuracy for both models, then the highest mean.

	Free-answer accuracy							
Model / mode	Raw	Summary	Summary−raw	Stated	Valid	Truncated	Forced	Median tokens
Qwen3-VL-8B-Instruct, direct	**??**	**??**	**??**	**??**	–	–	–	1
Qwen3-VL-8B-Thinking	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**

Table 16: Reasoning subset (development episodes; main task at trials 20, 40, 60, 80; controls at 40 and 80). Direct: the instruction-tuned checkpoint's single-token answer on the same items. Accuracy is over valid free answers; Valid, Truncated and Forced are proportions of all items.

accuracy by 7.4 and 9.6 points (main effects; 95% CIs [5.2, 9.5] and [7.6, 11.5]) and lowered the weight on the previous outcome from .48–.50 to .12, whereas replacing the reports had no effect on its own ($+1.1$ and -0.2 points). With both replaced, accuracy was 1.5–2.1 points below the correctness-coded history, so length and layout account for 14–20% of the difference between the raw history with correctness and the coded history, and the visible outcome label for the rest. The outcome label thus elicits the copying of the previous outcome even when who was correct is stated. Predictions for the held-out replication in five models were written before it was run. **[TODO: held-out replication]**

Prompt wording (exploratory, development episodes). We re-ran the four text history formats for Qwen3-VL-8B and InternVL3.5-8B with two alternative wordings; episodes, names, answer words, list order and readout were unchanged. Reworded sentences keep the original layout with different phrasing ("In trial 3, Sam said cat and Ken said dog; it turned out cat."; "Tally over the 3 previous trials: Ken was right 2 times and Sam was right 1 times."). A terse record uses one line per trial ("#3 | Sam=cat | Ken=dog | truth=cat"; "Correct counts so far (3 trials): Ken=2, Sam=1."). Table 22 gives the results. With reworded sentences, every result was preserved: raw history at .47–.49, summary−raw $+.12$ and $+.15$, the correctness-coded history at or above the summary, appending correctness recovering about 40% of the gap, and a positive weight on the previous outcome. The terse record changed the picture in both directions. Raw-history accuracy rose to .55–.56 and the weight on the previous outcome fell, whereas the correctness-coded and summarized formats fell to .56–.60, and InternVL3.5-8B followed the count summary on only 72% of non-tied trials (98% with the original wording). The summary−raw gap shrank to $+.03$ [.01, .06] (Qwen3-VL-8B) and $+.01$ [$-.02$, .05] (InternVL3.5-8B). Raw-history accuracy stayed below the count rule (.63) in every wording, but the size of the raw–summarized gap depends on how the history is laid out, and we interpret it only for the natural-language layout used in all main conditions. The terse record is also run for the three larger models. **[TODO: larger models, terse record]**

Which part of the terse record matters. The terse record changes three parts of the prompt: the task description, the history lines together with the count summary, and the current-trial line together with the question. We crossed the original and terse version of each part (2^3 prompts; the all-original prompt reproduces the runner's prompts byte for byte; Table 21). No single part accounts for the change. Each terse part, on its own, lowered the summary−raw gap by 0.3–5.4 points (main effects of 2.7–6.3 points), and the gap stayed positive with a 95% CI above zero in all six mixed prompts. The largest single changes were these: the terse question lowered accuracy with the correctness-coded history (main effect -3.6 and -4.3 points); the terse task description and count notation lowered agreement with the count rule (InternVL3.5-8B: -8.6 and -10.2 points); and for InternVL3.5-8B raw-history accuracy rose mainly when

		Median forgetting rate ϕ				Slope difference (model − human)	
Agent	Hazard info	$h = 0.1$	$h = 0.2$	$h = 0.4$	Slope	Est. [95% CI]	p
Humans ($n = 30$)	told	.079	.110	.255	0.47	–	–
Qwen3-VL-8B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??
InternVL3.5-8B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??
Gemma-3-27B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??
Qwen3-VL-32B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??
InternVL3.5-38B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??

Table 17: Psych-101 changing bandit (Xiong et al. 2023). Models are conditioned on each participant's history; ϕ is fitted to choices sampled from the model (20 draws, median). Slope: median over participants of the per-participant slope of ϕ on hazard rate (humans: our re-analysis). The paired test has power of about 0.5 and an upward bias of about $+0.09$ (calibration above), so it is descriptive only.

Model	Raw	Summary	Gap [95% CI]	p
Qwen3-VL-8B	??	??	??	??
InternVL3.5-8B	??	??	??	??
Gemma-3-27B	??	??	??	??
Qwen3-VL-32B	??	??	??	??
InternVL3.5-38B	??	??	??	??

Table 18: Primary comparison on the robustness set (24 episodes with independent outcomes and irregular reversals).

both the history lines and the current trial used the same one-line record (.477 and .491 with either alone, .559 with both). This last pattern fits the outcome-evaluation account: the gain under the raw history appeared when each past trial's reports and outcome were aligned in the same layout as the current reports, which makes deciding who was correct easier.

K. Models, Revisions and Compute

Table 23 lists the models and Table 24 the Hugging Face checkpoints and pinned revisions. A 72B model was planned as a scale extension and larger reasoning models as a supplement, but neither was run for compute reasons.

Image preprocessing. Icons and photos are rendered at 224 px for Qwen models and at 448 px (one tile, no tiling) for InternVL3.5. For Gemma-3, 224-px renders are resized by the processor to the fixed 896-px input (256 image tokens; pan-and-scan off). The correctness-coded chart is rendered at 448 px for all models.

Software and hardware. All reported results on held-out episodes, the robustness set and Psych-101, the control conditions, and the development runs used to select the window length were run on NVIDIA H200 GPUs (141 GB), one GPU per run, with PyTorch 2.8.0 (CUDA 12.8) and Transformers 4.57.1. The exploratory development checks (icon selection, prompt wording, masking) were run on NVIDIA RTX 3090 GPUs with the same model revisions, precision and readout. Each run stores the rendered prompt hash, the raw candidate logits, the probability mass on the two answer tokens, and a fingerprint of the configuration. GPU-hours are in Table 25.

Readout quality. Because y_t is normalized over the two candidate tokens only, we report for every model, format and condition the probability mass on the two candidates (mean, minimum and share of trials below 0.5) and the format-following rate (share of trials on which the greedy output is one of the two answer words); see Table 25.

L. Pilot Results

Before the analysis plan was fixed, we ran a pilot on RTX 3090 GPUs with an earlier version of the episodes (16 episodes), whose instructions always introduced the advisors in the same order; this confound was removed in the episodes used here, so the pilot only motivated the design. In Qwen3-VL-32B, the change-point summary exceeded the raw history by $+.191$ [.161, .220] and the all-trial summary by $+.145$; the model given a summary matched the rule alone ($+.001$); under raw histories its choices agreed most with repeating the previous outcome ($\kappa = 0.25$) and its integration kernel was nearly flat. With photo reports it followed the advisor listed first (1.00 when advisor A's photo came first, .03 when second), and Qwen3-VL-8B read single cells of the correctness chart on .56 of items. These observations shaped the control conditions, the binding checks and the chart gate; none of the pilot data enter the results.

	Text cover stories							
Model	Color	Weather	Stock	Sports	Medical	News	Sign test p	Pet (photo)
Qwen3-VL-8B	??	??	??	??	??	??	??	??
InternVL3.5-8B	??	??	??	??	??	??	??	??
Gemma-3-27B	??	??	??	??	??	??	??	??
Qwen3-VL-32B	??	??	??	??	??	??	??	??
InternVL3.5-38B	??	??	??	??	??	??	??	??

Table 19: Summary−raw gap by cover story (held-out; eight episodes per text cover story). Pet uses photo reports in its own episode slot and is not part of the sign test.

	Raw + correctness		Reports shown		Outcome shown		Both replaced		Coded	
Model	Acc.	Copy	Acc.	Copy	Acc.	Copy	Acc.	Copy	Acc.	Copy
Development episodes										
Qwen3-VL-8B	.534	.50	.576	.12	.513	.48	.620	.04	.640	.03
InternVL3.5-8B	.530	.48	.602	.12	.504	.27	.625	.02	.640	.01
Held-out episodes										
Qwen3-VL-8B	??	??	??	??	??	??	??	??	??	??
InternVL3.5-8B	??	??	??	??	??	??	??	??	??	??
Gemma-3-27B	??	??	??	??	??	??	??	??	??	??
Qwen3-VL-32B	??	??	??	??	??	??	??	??	??	??
InternVL3.5-38B	??	??	??	??	??	??	??	??	??	??

Table 20: Masking parts of the raw history with correctness (text reports; twin-averaged accuracy, $t \geq 2$). Reports shown: outcome label replaced; Outcome shown: reports replaced; Both replaced: same information as the coded history. Copy: regression weight on repeating the previous outcome. Development rows are exploratory (RTX 3090); held-out predictions were fixed beforehand: replacing the outcome label raises accuracy, replacing both beats the raw history with correctness, and replacing the outcome label matters more than replacing the reports.

			Qwen3-VL-8B				InternVL3.5-8B			
Task	History	Question	Raw	Coded	Summ.	Summ.−raw [95% CI]	Raw	Coded	Summ.	Summ.−raw [95% CI]
O	O	O	.485	.640	.617	+.132 [.108, .153]	.471	.640	.628	+.157 [.140, .174]
O	O	T	.512	.613	.627	+.115 [.093, .137]	.491	.604	.617	+.126 [.106, .146]
O	T	O	.494	.633	.623	+.129 [.114, .145]	.477	.637	.600	+.122 [.099, .145]
O	T	T	.532	.588	.608	+.076 [.051, .105]	.559	.616	.597	+.038 [.007, .065]
T	O	O	.506	.637	.600	+.094 [.066, .121]	.507	.634	.610	+.103 [.073, .132]
T	O	T	.520	.599	.600	+.080 [.052, .108]	.493	.562	.600	+.107 [.079, .131]
T	T	O	.526	.616	.600	+.074 [.040, .104]	.515	.634	.583	+.068 [.035, .100]
T	T	T	.563	.580	.597	+.034 [.008, .059]	.550	.593	.564	+.014 [−.019, .045]

Table 21: Decomposition of the terse record (development episodes, text reports; twin-averaged accuracy, $t \geq 2$). O: original wording of that part; T: terse version. Task: task description; History: history lines and count summary; Question: current-trial line and question. The first row is the original prompt and the last row the terse record of Table 22.

Model	Wording	Raw	Raw+c.	Coded	Summ.	Summ.−raw [95% CI]	Copy
Qwen3-VL-8B	Original	.485	.534	.640	.617	+.132 [.108, .153]	.46
	Reworded	.488	.533	.624	.607	+.119 [.093, .144]	.50
	Terse	.563	.561	.580	.597	+.034 [.008, .059]	.32
InternVL3.5-8B	Original	.471	.530	.640	.628	+.157 [.140, .174]	.28
	Reworded	.473	.538	.644	.627	+.153 [.133, .175]	.26
	Terse	.550	.559	.593	.564	+.014 [−.019, .045]	.13

Table 22: Prompt-wording check (development episodes, text reports; accuracy twin-averaged, $t \geq 2$). Copy: regression weight on the previous outcome under the raw history.

Model	LM family	Params	Image input
Qwen3-VL-8B	Qwen3	8B	224 px
InternVL3.5-8B	Qwen3	8B	448 px
Gemma-3-27B	Gemma 3	27B	896 px[a]
Qwen3-VL-32B	Qwen3	32B	224 px
InternVL3.5-38B	Qwen3	38B	448 px
Qwen3-VL-8B-Thinking[b]	Qwen3	8B	224 px

Table 23: Models (frozen; pinned revisions in Table 24). Top block: core models (instruction-tuned). InternVL3.5: one 448-px tile. [a]224-px renders resized by the processor. [b]Reasoning subset only.

Model	Checkpoint	Revision
Qwen3-VL-8B	Qwen/Qwen3-VL-8B-Instruct	0c351dd01ed87e9c1b53cbc748cba10e6187ff3b
InternVL3.5-8B	OpenGVLab/InternVL3_5-8B-HF	741a7d03020411e666c6109218ab71e08151ef86
Gemma-3-27B	google/gemma-3-27b-it	005ad3404e59d6023443cb575daa05336842228a
Qwen3-VL-32B	Qwen/Qwen3-VL-32B-Instruct	0cfaf48183f594c314753d30a4c4974bc75f3ccb
InternVL3.5-38B	OpenGVLab/InternVL3_5-38B-HF	7c830fc25e874f9a861087e3c70587f2b9b555f3
Qwen3-VL-8B-Thinking	Qwen/Qwen3-VL-8B-Thinking	92f3c4b4feadd3a016ef468d103bb5f58b2a2c6b

Table 24: Checkpoints and pinned revisions (resolved 2026-11-11).

	Candidate probability mass			Format	
Model	Mean	Min	Share < 0.5	following	GPU-hours
Qwen3-VL-8B	1.000	.997	.000	1.000	**??**
InternVL3.5-8B	.996	.112	.002	.996	**??**
Gemma-3-27B	**??**	**??**	**??**	**??**	**??**
Qwen3-VL-32B	.996	.907	.000	1.000	**??**
InternVL3.5-38B	**??**	**??**	**??**	**??**	**??**
Qwen3-VL-8B-Thinking	**??**	**??**	**??**	**??**	**??**
Total					**??**

Table 25: Readout quality pooled over formats and conditions per model (per cell in the released analysis output). **[DRAFT: development episodes (16) and four models; replace with held-out results]** (main task, four history formats).